Titanic Passenger Survival

Programming for Data Science Assignment

What the analysis shows

This analysis uses passenger records to compare survival with age, gender, ticket class, and fare. Overall, 38.4% of the 891 passengers in the supplied file survived. Survival differed strongly by gender and class: women and passengers in higher classes had higher observed survival rates. These are descriptive patterns in this dataset, not proof that any one factor caused survival.

Dataset and method

The CSV contains 891 passenger rows and 12 columns. Survival is coded 1 for survived and 0 for did not survive. I excluded records with missing age only from age-based summaries; 177 ages are missing (19.9%). The age chart uses six bands so rates can be compared clearly. The fare chart plots individual passengers, uses a logarithmic fare axis to show both typical and very high fares, and encodes age with marker size. Small horizontal jitter separates passengers in the same class.

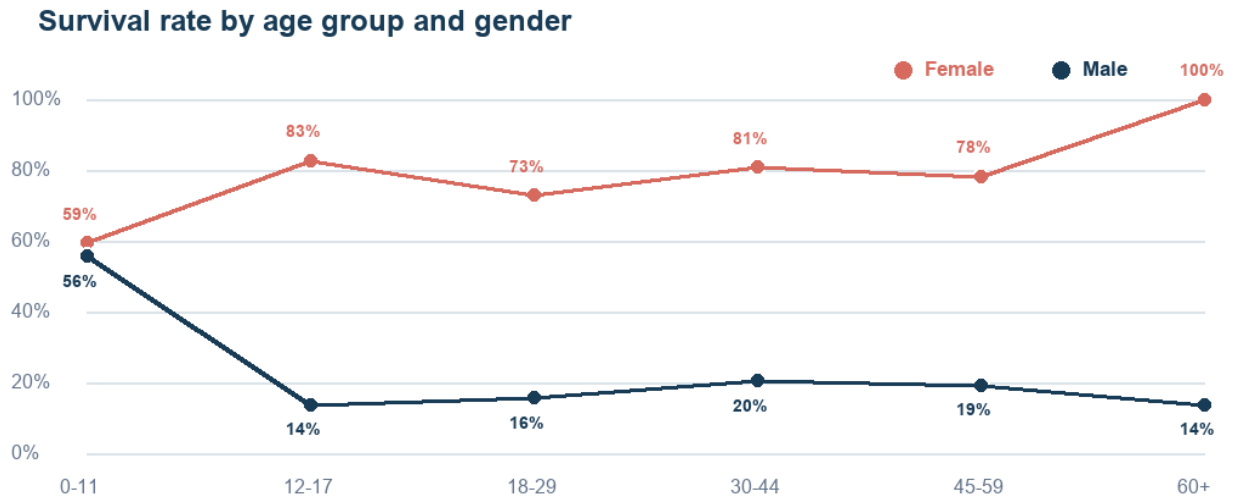
Measure	Observed result
Overall survived	38.4%
Female survived	74.2% (n=314)
Male survived	18.9% (n=577)
1st / 2nd / 3rd class	63.0% / 47.3% / 24.2%

How to read the rates: Each rate is the number who survived divided by the number of passengers in that group. Passenger class 1 is the highest fare/status class and class 3 the lowest.

The assignment asks for three visual comparisons and an interpretation of which passenger groups were more likely to survive. The following pages show those results.

1. Survival and age

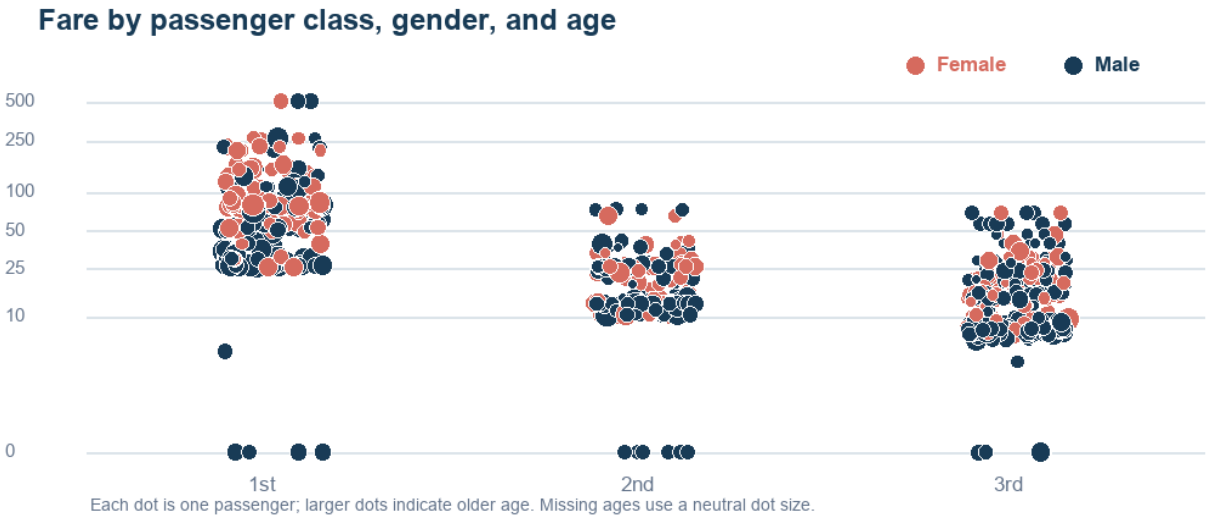
Survival percentage by age band, split by gender. The labels show group rates; each age band includes only passengers whose age was recorded.



Across most age bands, female passengers had higher survival rates than male passengers. Children under 12 were the exception to a large gender gap: rates were similar (59% for girls and 56% for boys). The 60+ female rate is based on only four passengers, so it should be read cautiously.

2. Fare and passenger class

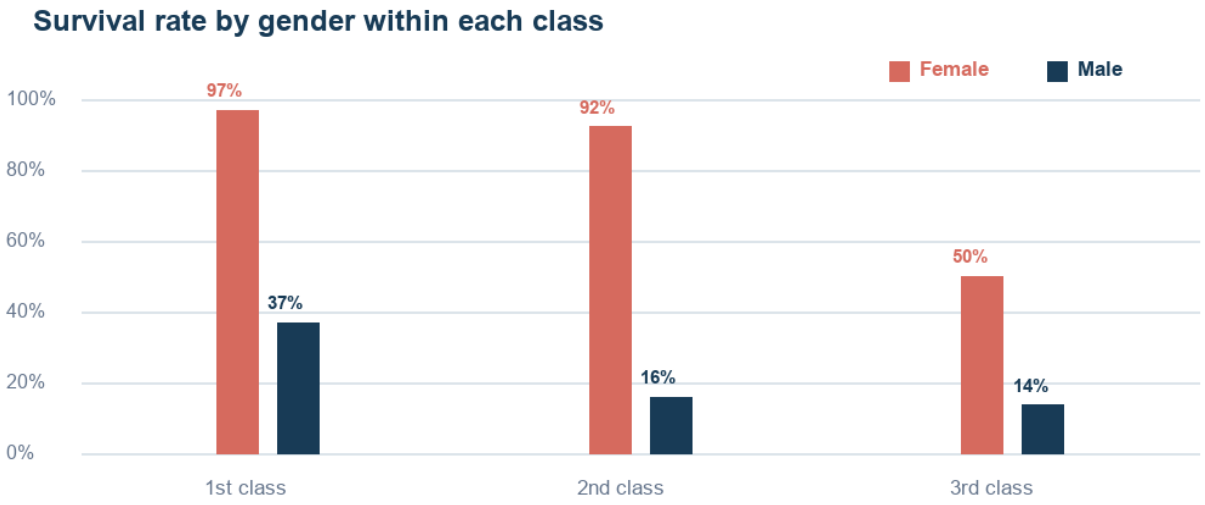
Each point is one passenger. Color identifies gender and marker size encodes age; larger points indicate older passengers. The logarithmic vertical scale keeps low and high fares visible together.



Fares are generally higher in first class and lower in third class. Median fares were 60.29, 14.25, and 8.05 fare units for classes 1, 2, and 3, respectively. There is substantial variation within each class, including a few unusually high fares.

3. Survival by gender and class

Bars compare survival rates for female and male passengers within each passenger class.



Class	Female	Male
1 (first class)	96.8%	36.9%
2 (second class)	92.1%	15.7%
3 (third class)	50.0%	13.5%

4. Who was more likely to survive?

Female passengers were much more likely to survive than male passengers overall (74.2% versus 18.9%). The same pattern appears within each class: female survival rates were 96.8% in first class, 92.1% in second, and 50.0% in third. Male rates were lower in every class. This shows that the gender difference is present even when comparing passengers within the same class.

Class also mattered in the observed data. Survival was 63.0% in first class, 47.3% in second, and 24.2% in third. Age alone was less decisive in this summary: the mean recorded age was 28.3 years among survivors and 30.6 among non-survivors. Family group size showed a mixed pattern: passengers traveling with one or two listed relatives had a 56.3% survival rate, compared with 30.4% for passengers traveling alone and 34.1% for those with three or more.

Conclusion

In this passenger file, being female and traveling in a higher class were the clearest markers associated with survival. Age and family size add context, but their patterns were less consistent. Missing ages and the uneven sizes of passenger groups limit some comparisons; these findings describe the recorded passengers and should not be generalized as causal effects.

Data note: percentages are calculated from the supplied Titanic-Dataset.csv (891 records). Age-based results exclude passengers with missing ages; chart labels are rounded.